Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Initial momentum is 0.3 kg·m/s and measured final momentum is 0.29 kg·m/s. Find percent deviation. Explain why this is not itself a measured uncertainty.

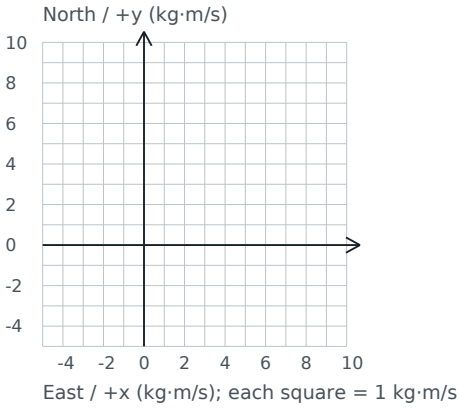
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

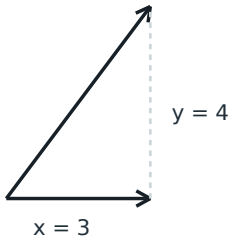
Your turn: Draw a tip-to -tail momentum diagram and label each axis with units.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Two incoming arrows point east and north. Sketch the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

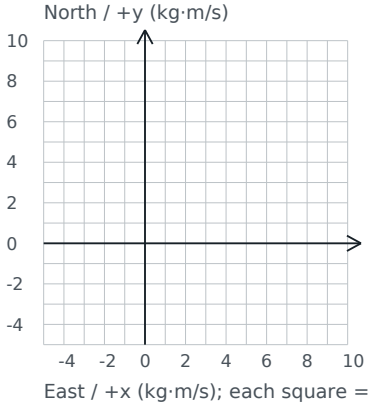


The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart at 3 m/s east sticks to a 1 kg cart at 4 m/s north. Find final velocity components, speed and angle north of east.



Marzano check - circle: ① ② ③

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Continue down each column.

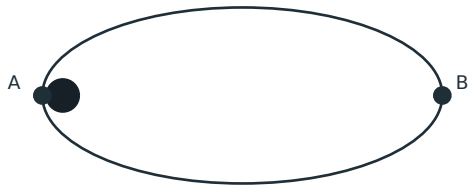
05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

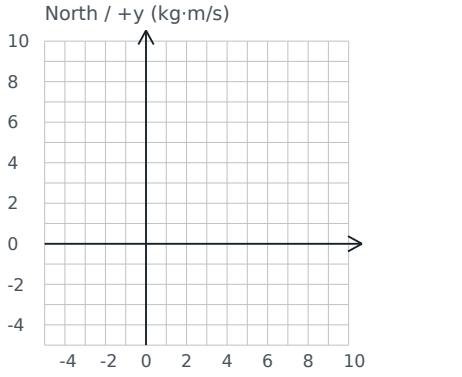
Two ideal Sun-orbits have $a_B = 4\ a_A$. Find T_B/T_A. Explain why the same normalized Sun-orbit equation is not automatically valid around another body.
Label the orbital features used in your explanation.



Marzano check - circle: 1 2 3

07 Exit check

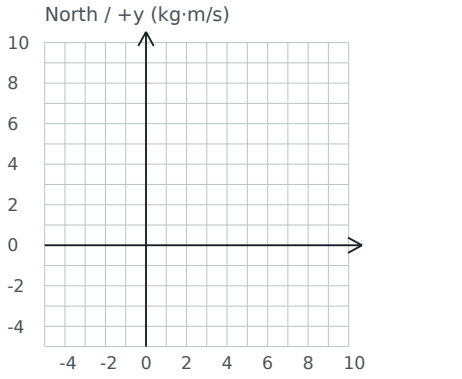
Initial total momentum is (10, 0) kg·m/s. One fragment leaves with (4, 3). Find the other momentum vector and its magnitude.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two pucks stick on an ideal air table. One has mass 1 kg and moves east at 4 m/s. The other has mass 3 kg and moves north at 2 m/s. Draw the momentum sum. Find the joined pair's velocity components, speed and direction.



Marzano check - circle: 1 2 3